

Project Name: LifeLineX – Organ Transport & Allocation System

Phase 4: Process Automation (Admin)

Objective

The goal of this phase is to introduce automation into the LifeLineX system. By using Salesforce automation tools such as Validation Rules, Workflow Rules, Process Builder, Approval Processes, Flows, and Notifications, the system ensures accurate data entry, efficient process execution, and real-time communication with all stakeholders.

1. Validation Rules

Validation Rules ensure that critical data is captured before a record can be saved.

- **Organ Object**
 - Rule: Condition field must not be blank.
 - Formula:
 - ISBLANK(Condition__c)
 - Error Message: “Organ Condition must be specified.”

The screenshot displays the Salesforce Setup interface for configuring a Validation Rule on the 'Organ' object. The left sidebar shows the navigation menu with 'Setup' and 'Object Manager' selected. The main content area is titled 'Organ Validation Rule' and includes a description of validation rules. The 'Validation Rule Edit' section shows the rule name 'Expiry_must_be_future', which is active. The 'Error Condition Formula' section contains the formula: `AND( NOT( ISBLANK( Expiry_DateTime__c ) ), Expiry_DateTime__c < NOW() )`. The 'Error Message' section shows the message: 'Expiry dateTime must be in the future.' The 'Error Location' is set to 'Field' and 'Expiry DateTime'.

- **Transport Case Object**
 - Rule: Courier must be assigned before saving a transport case.

Setup Home Object Manager

Search Setup

Setup > OBJECT MANAGER

Transport Case

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Transport Case Validation Rule

Define a validation rule by specifying an error condition and a corresponding error message. The error condition is written as a Boolean formula expression that returns true or false. When the formula expression returns true, the save will be aborted and the error message will be displayed. The user can correct the error and try again.

Validation Rule Edit Save Save & New Cancel

Rule Name

Active ☒

Description

Error Condition Formula f Required Information

Example: `Discount_Percent__c > 30` More Examples...
Display an error if Discount is more than 30%
If this formula expression is **true**, display the text defined in the Error Message area

```
AND(
  NOT(ISBLANK(Pickup_Time__c)),
  NOT(ISBLANK(Delivery_Time__c)),
  Delivery_Time__c < Pickup_Time__c
)
```

No errors found

Functions
-- All Function Categories --
CASE
CASESAFEID
CEILING
CHR
CONTAINS
COS

ABS(number)
Returns the absolute value of a number, a number without its sign
[Help on this function](#)

Error Message

Example: `Discount percent cannot exceed 30%`
This message will appear when Error Condition formula is **true**

Error Message

This error message can either appear at the top of the page or below a specific field on the page

Error Location ☐ Top of Page ☒ Field

Setup Home Object Manager

Search Setup

Setup

Quick Find

Setup Home

Salesforce Go

Service Setup Assistant

Commerce Setup Assistant

Field Service Setup Home (Beta)

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

Salesforce Go

Service Setup Assistant

Commerce Setup Assistant

Field Service Setup Home (Beta)

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

Optimizer

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

> Email

PLATFORM TOOLS

> Subscription Management

Transport Case Validation Rule

Define a validation rule by specifying an error condition and a corresponding error message. The error condition is written as a Boolean formula expression that returns true or false. When the formula expression returns true, the save will be aborted and the error message will be displayed. The user can correct the error and try again.

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Example: `Discount_Percent__c > 30` More Examples...
Display an error if Discount is more than 30%
If this formula expression is **true**, display the text defined in the Error Message area

```
AND(
  ISPICTHROW(Status__c, "Delivered"),
  ISBLANK(Delivery_Time__c)
)
```

No errors found

Functions
-- All Function Categories --
CONTAINS
COS
CURRENCYRATE
DATE
DATEVALUE
DATEVALUE

ABS(number)
Returns the absolute value of a number, a number without its sign
[Help on this function](#)

Error Message

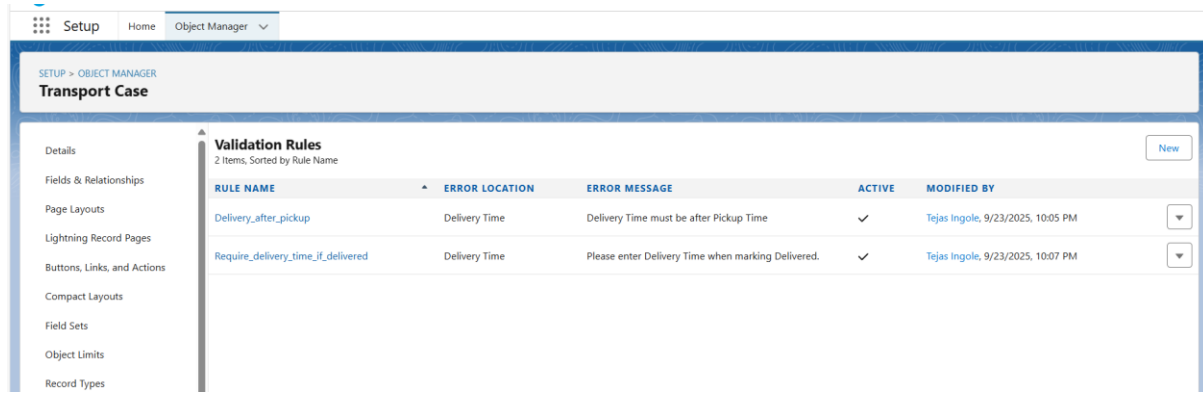
Example: `Discount percent cannot exceed 30%`
This message will appear when Error Condition formula is **true**

Error Message

This error message can either appear at the top of the page or below a specific field on the page

Error Location ☐ Top of Page ☒ Field

Purpose: Prevents incomplete or invalid records.

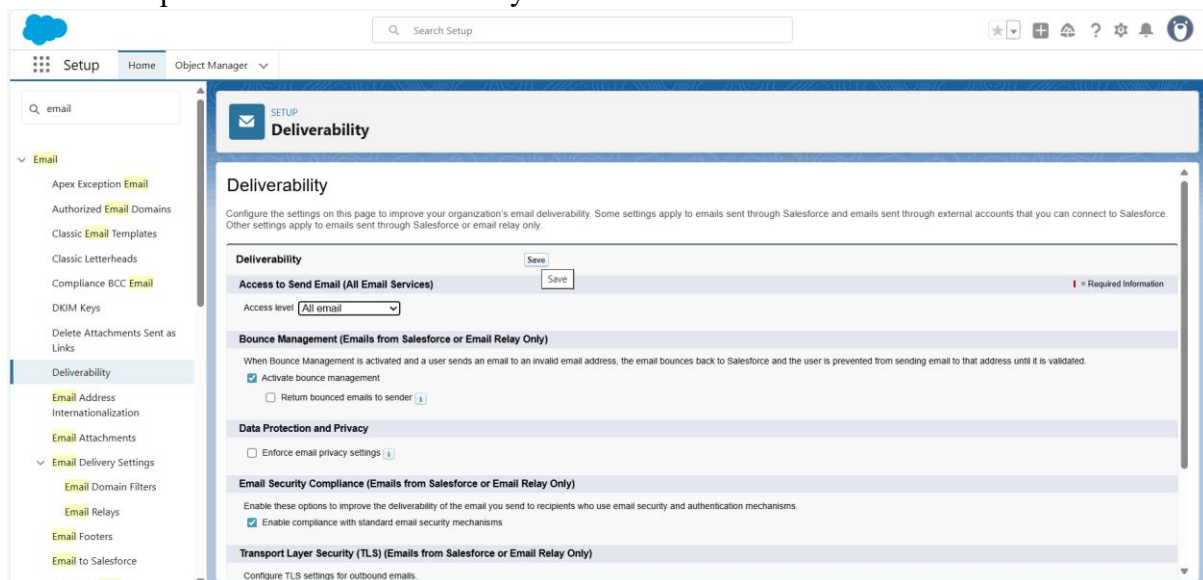


Email templates & deliverability (prereq for alerts)

Create templates to re-use in flows and alerts.

Enable Email Deliverability (admin)

1. Setup → Email → Deliverability → Set Access level to All email → Save.



Create Email Template (Lightning)

1. Setup → Quick Find → Email Templates → New Email Template
2. Folder: Unfiled Public Templates (or create project folder)
3. Name: TransportCase_Status_Notification
4. Developer Name auto-filled.
5. Subject: LifeLineX: Transport Case {!Transport_Case__c.Case_Number__c} -
 {!Transport_Case__c.Status__c}

Setup

Home

Object Manager

email te

Email

Classic Email Templates

Lightning Email Templates

Didn't find what you're looking for? Try using Global Search.

Classic Email Templates

Custom Email Template

New Template

Select the appropriate folder and enter the name and description for the email template below. Note that the Description field is for internal use only.

Once you have completed the HTML email template check the "Available For Use" box to make this template available to your users.

Step 2. Custom Email Template: New Template

Step 2 of 4

Previous

Next

Cancel

Email Template Information

Required Information

Folder

Untitled Public Classic Email Templates

Available For Use

Email Template Name

TransportCase_Status_Not

Template Unique Name

TransportCase_Status_Not

Encoding

Unicode (UTF-8)

Description

Subject

LifeLineX: Transport Case (Transport_Case__c Case_Number__c) - (Transp

Previous

Next

Cancel

email te

Email

Classic Email Templates

Lightning Email Templates

Didn't find what you're looking for? Try using Global Search.

Classic Email Templates

Contact Fields

Copy and paste the merge field value into your HTML content below:

Step 3. Create HTML version

Step 3 of 4

Previous

Previous

Next

Cancel

HTML Email Content

Required Information

Subject

LifeLineX: Transport Case (Transport_Case__c Case_Number__c) - (Transp

HTML Body

Hello (User.Name).

Transport Case (Transport_Case__c Case_Number__c) for Organ

(Transport_Case__c Organ__r Organ_Type__c) is now (Transport_Case__c Status__c)

Pickup: (Transport_Case__c Pickup_Time__c)

ETA: (Transport_Case__c ETA__c)

Regards,

LifeLineX

Setup

Home

Object Manager

email te

Email

Classic Email Templates

Lightning Email Templates

Didn't find what you're looking for? Try using Global Search.

Classic Email Templates

Custom Email Template

TransportCase_Status_Notification

Preview your email template below.

Email Template Detail

Edit Properties

Edit HTML Version

Edit Text Version

Delete

Clone

Email Templates from Salesforce

Untitled Public Classic Email Templates

Available For Use

Email Template Name

TransportCase_Status_Notification

Last Used Date

Template Unique Name

TransportCase_Status_Notification_send

Times Used

Encoding

Unicode (UTF-8)

Author

Tejas Ingole (Change)

Description

Created By

Tejas Ingole, 9/23/2025, 10:22 PM

Modified By

Tejas Ingole, 9/23/2025, 10:22 PM

Edit Properties

Edit HTML Version

Edit Text Version

Delete

Clone

Email Template

Send Test and Verify Merge Fields

Subject

LifeLineX: Transport Case (Transport_Case__c Case_Number__c) - (Transport_Case__c Status__c)

HTML Preview

Hello (User.Name), Transport Case (Transport_Case__c Case_Number__c) for Organ

(Transport_Case__c Organ__r Organ_Type__c) is now (Transport_Case__c Status__c). Pickup:

(Transport_Case__c Pickup_Time__c) ETA: (Transport_Case__c ETA__c) Regards, LifeLineX

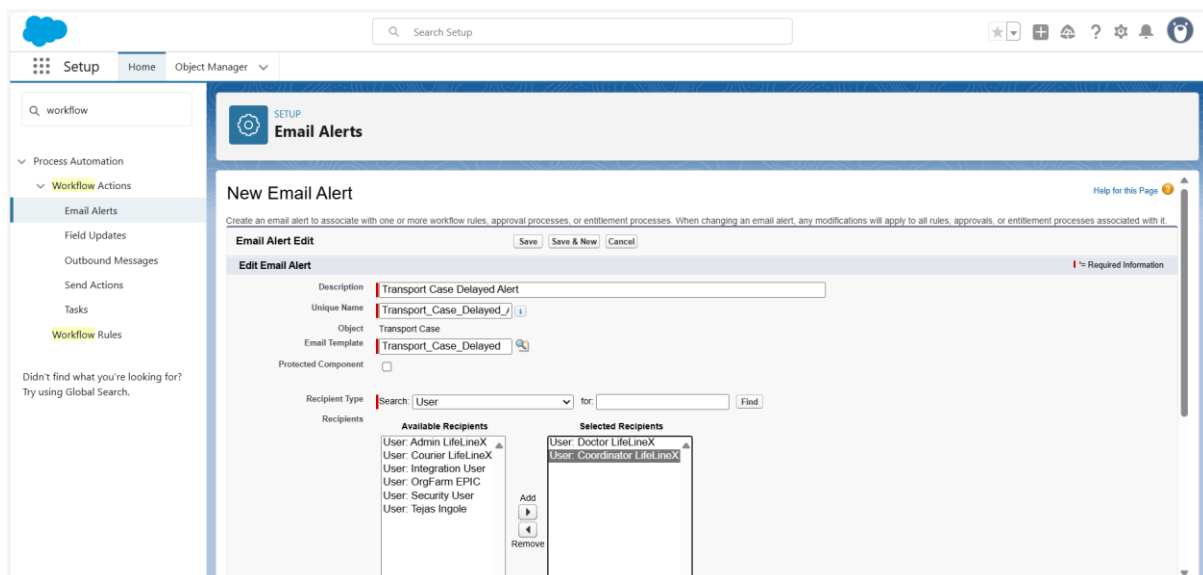
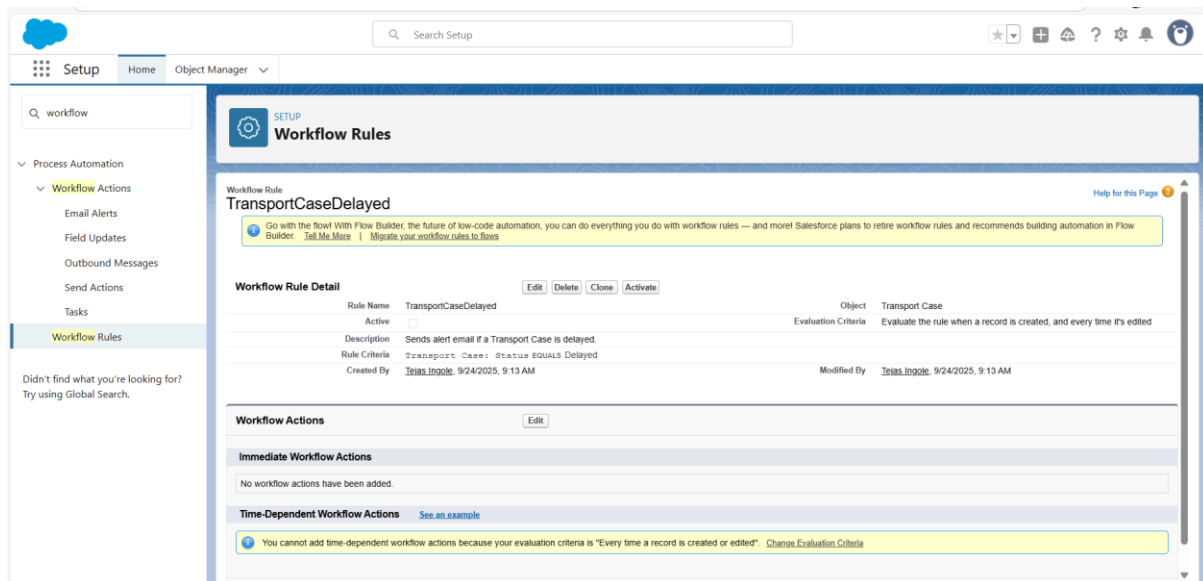
2. Workflow Rules

Workflow automates simple tasks like sending emails or field updates.

- **Transport Case Workflow**
 - Criteria: Status = "Delayed".
 - Action: Send Email Alert → Recipient: Assigned Doctor / Coordinator.
 - Email Template: "Transport Case Delayed".

The screenshot shows the Salesforce Classic Email Templates setup page. The left sidebar has a search bar with 'email tem' and a list of categories: Email, Classic Email Templates (selected), and Lightning Email Templates. The main content area is titled 'Classic Email Templates' and shows the details for a 'Text Email Template' named 'Transport_Case_Delayed'. The 'Email Template Detail' section includes fields for Email Template Name, Template Unique Name, Encoding, Author, Description, Created By, and Modified By. The 'Email Template' section shows the subject and plain text preview. The subject is 'Transport Case ({Transport_Case__c Case_Number__c}) is Delayed'. The plain text preview starts with 'Hello ({Transport_Case__c Doctor__r Name}).' and mentions that the transport case has been delayed.

The screenshot shows the Salesforce Workflow Rules setup page. The left sidebar has a search bar with 'workflow' and a list of categories: Process Automation, Workflow Actions, Email Alerts, Field Updates, Outbound Messages, Send Actions, Tasks, and Workflow Rules (selected). The main content area is titled 'Workflow Rules' and shows the 'New Workflow Rule' setup for 'Transport Case'. The 'Edit Rule' section includes fields for Object, Rule Name, and Description. The 'Evaluation Criteria' section shows options for when to evaluate the rule. The 'Rule Criteria' section shows a table with fields, operators, and values. The first row is 'Transport Case: Status' with the operator 'equals' and the value 'Delayed'. The table is followed by 'AND' and 'Add Filter Logic...' buttons.



Purpose: Alerts stakeholders when a critical case is delayed.

3. Process Builder

Process Builder automates record updates and related actions.

- **Organ Assignment Process**

- Trigger: When Organ status = "Ready".
- Action: Update related Transport Case → Status = "Pending Pickup".

Purpose: Ensures that transport arrangements start as soon as an organ is ready.

Setup

Home

Object Manager

Search Setup

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with Process Builder—and more! Salesforce plans to retire Process Builder and recommends building automation in Flow Builder.

Try Flow Builder | Use Migrate to Flow Tool

Process Builder

My Processes

Welcome to the Process Builder!

With the Process Builder you can easily build automation like approvals and follow-up emails, renewals and new-hire onboarding.

Learn More

Using the Process Builder (document)

New Process

Organ Assignment Process

Organ_Assignment_Process

Description

Updates Transport Case status when Organ is ready.

The process starts when *

A record changes

Cancel

Save

Setup

Home

Object Manager

Search Setup

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with Process Builder—and more! Salesforce plans to retire Process Builder and recommends building automation in Flow Builder.

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Process Builder - Organ Assignment Process

Expand All

Collapse All

View All Processes

Clone

Edit Properties

START

+ Add Object

+ Add Criteria

TRUE

IMMEDIATE ACTIONS

+ Add Action

TIME

SCHEDULED ACTIONS

Set Schedule

+ Add Action

STOP

Choose Object and Specify When to Start the Process

Object *

Organ

Start the process *

☐ only when a record is created

☒ when a record is created or edited

Advanced

Save

Cancel

Expand All

Collapse All

View All Processes

Clone

Edit Properties

START

Organ

+ Add Criteria

TRUE

IMMEDIATE ACTIONS

+ Add Action

STOP

Define Criteria for this Action Group

Criteria Name *

Organ Ready Criteria

Criteria for Executing Actions *

☒ Conditions are met

☐ Formula evaluates to true

☐ No criteria—just execute the actions!

Set Conditions

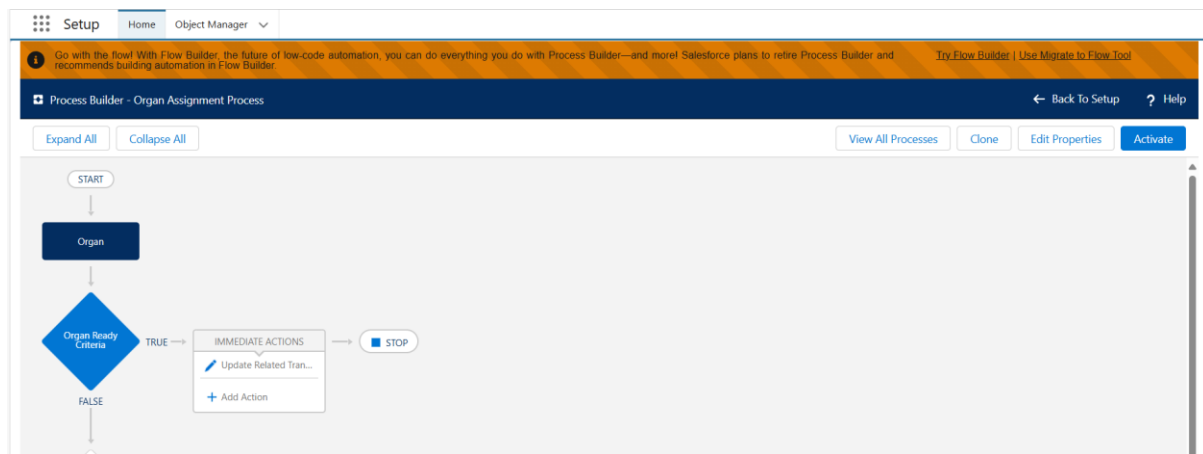
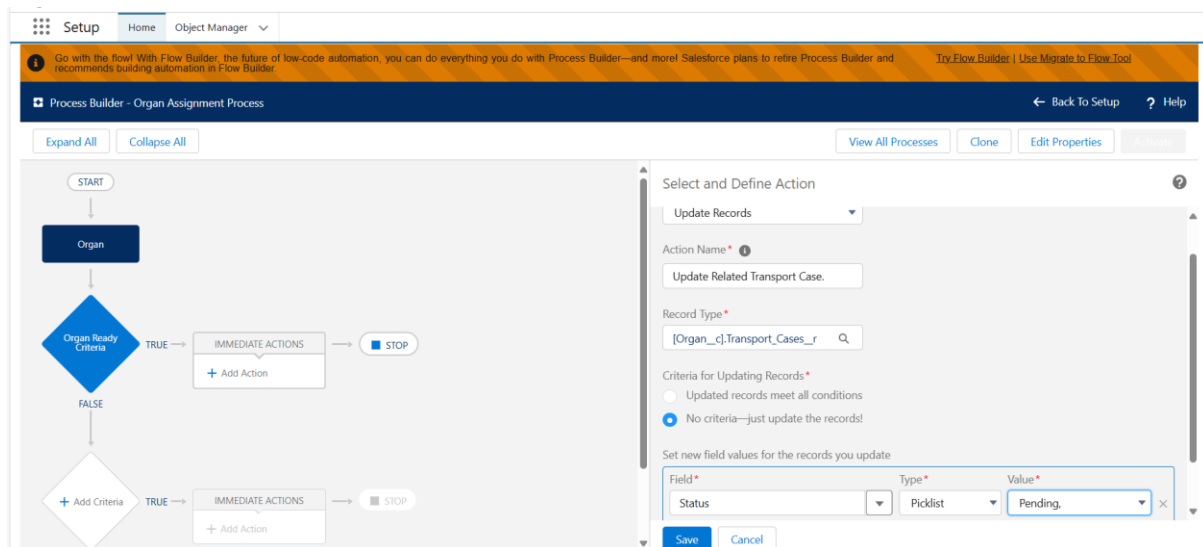
Field *	Operator *	Type *	Value *	
1	[Organ_c].Ava...Q	Equals	Picklist	Available, ...
+ Add Row				

Conditions *

☒ All of the conditions are met (AND)

Save

Cancel



4. Approval Process

Approvals ensure accountability before final actions.

- **Organ Approval Process**

- Trigger: When Organ status = "Ready".
- Approver: Doctor's Manager (or Admin).
- Final Approval Action: Update Organ status → "Approved".
- Final Rejection Action: Update Organ status → "Rejected".

Purpose: Ensures that only authorized organs are cleared for transplant.

Approval Processes

Approval Process Jump Start Wizard
Organ

Approval Process Information

The Jump Start wizard creates a one-step approval process for you in just a few minutes.
Enter a name for your process in the box below and then select an email template to notify the approver (optional).

Name: **Organ Approval Process**
Unique Name: **Organ_Approval_Process**
Approval Assignment Email Template: **Approves Organ before it**

Specify Entry Criteria

Use this approval process if the following criteria are met:

Field	Operator	Value
Organ: Availability Status	equals	Available
--None--	--None--	
--None--	--None--	
--None--	--None--	
--None--	--None--	

Select Approver

Using the options below, specify the user to whom the approval request should be assigned.

☐ Let the submitter choose the approver manually.
☐ Automatically assign an approver using a standard or custom hierarchy field: --None--
☒ Automatically assign to queue: **Organ Approvals Queue**
☐ Automatically assign to approver(s).

5. Flow Builder

Flows handle complex automation beyond Workflow/Process Builder.

- **Transport Case Alert Flow (Record-Triggered)**
 - Trigger: When Transport Case status = "In Transit".
 - Action: Create Alert record → "Organ is in transit".
 - Linked Transport Case = Current Case ID.

Purpose: Automatically creates alerts when transport updates occur.

Flow Builder

Record-Triggered Flow
Start
Object: **Transport Case**
Trigger: **A record is updated**
Conditions: **1**
Optimize for: **Fast Field Updates**
[Open Flow Trigger Explorer for Transp...](#)

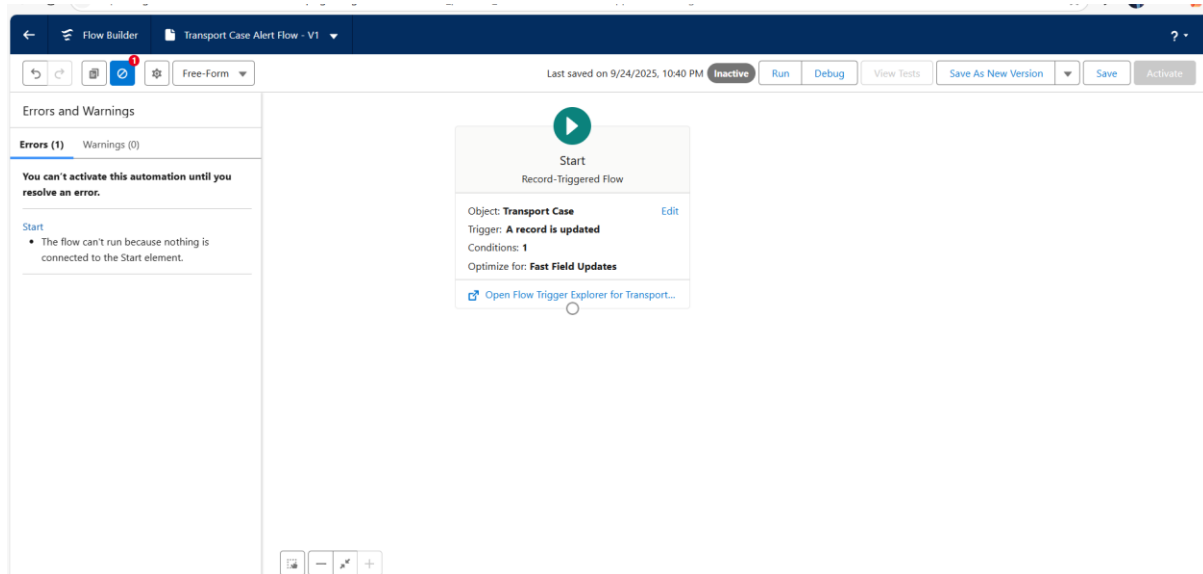
End

Configure Start

Select Object
Select the object whose records trigger the flow when they're created, updated, or deleted.
* Object: **Transport Case**

Configure Trigger
Trigger the Flow When:
☐ A record is created
☒ A record is updated
☐ A record is created or updated
☐ A record is deleted

Set Entry Conditions
Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.
If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.
Condition Requirements: **All Conditions Are Met (AND)**



6. Custom Notifications

Custom Notifications send real-time updates directly to users' desktop or mobile devices.

- **Notification: Organ Transit Update**
 - Triggered via Flow (Step 5).
 - Title: "Organ Transit Update".
 - Body: "Organ is now in transit."
 - Recipient: Assigned Doctor.

Purpose: Doctors and coordinators stay updated instantly.

